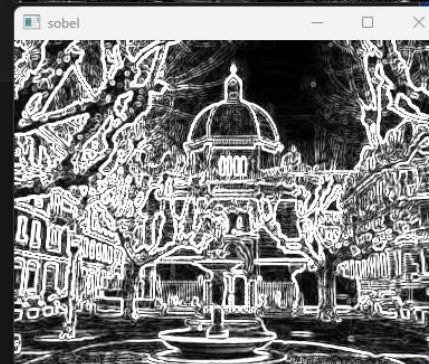
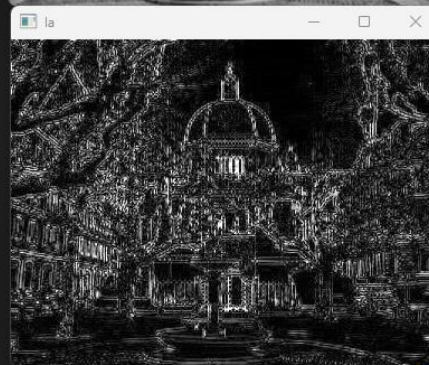
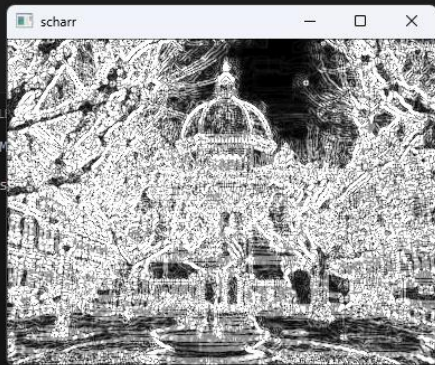


```

p1.py  p2.py  X
p2.py > ...
1  import cv2
2  import numpy as np
3
4  src = cv2.imread('geneva.jpg', cv2.IMREAD_GRAYSCALE)
5  rows, cols = src.shape
6
7  sobelx = cv2.Sobel(src, cv2.CV_32F, 1, 0, ksize=3)
8  sobely = cv2.Sobel(src, cv2.CV_32F, 0, 1, ksize=3)
9  dst_opencv = cv2.magnitude(sobelx, sobely)
10 dst_opencv = np.uint8(np.clip(dst_opencv, 0, 255))
11
12 dstx = cv2.Scharr(src, cv2.CV_32F, 1, 0) # 計算 x 軸影像梯度
13 dsty = cv2.Scharr(src, cv2.CV_32F, 0, 1) # 計算 y 軸影像梯度
14 dstx = cv2.convertScaleAbs(dstx) # 將負值轉正值
15 dsty = cv2.convertScaleAbs(dsty) # 將負值轉正值
16 dst_scharr = cv2.addWeighted(dstx, 0.5, dsty, 0.5, 0)
17
18 dstla = cv2.Laplacian(src, cv2.CV_64F, ksize=3, scale=0.5)
19 dstla = cv2.convertScaleAbs(dstla)
20
21 cv2.imshow("src", src)
22 cv2.imshow("scharr", dst_scharr)
23 cv2.imshow("sobel", dst_opencv)
24 cv2.imshow("la", dstla)
25 cv2.waitKey(0)
26 cv2.destroyAllWindows()

```



PROBL

(c) H

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\\week7\p2.py"